

Iris Flower Classification Report

Objective

The objective of this project is to develop a machine learning classifier capable of identifying Iris flower species using flower measurements.

Dataset Description

The Iris dataset contains 150 flower samples belonging to three species:

- Setosa
- Versicolor
- Virginica

Each sample contains four measurements:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Data Preparation

The dataset was loaded from scikit-learn. Features and target labels were separated for training and evaluation.

Features:

- Flower measurements

Target:

- Species label

Train-Test Split

The dataset was divided into:

- 80% Training Data
- 20% Testing Data

A `random_state` value of 42 was used to ensure reproducible results.

Model Training

A Decision Tree Classifier was trained using the training dataset.

The model learns decision rules based on feature values and predicts the corresponding flower species.

Model Evaluation

Model performance was evaluated using:

- Accuracy Score
- Confusion Matrix

Accuracy indicates the percentage of correct predictions on unseen test data.

Confusion Matrix Analysis

The confusion matrix compares actual classes with predicted classes.

- Diagonal values represent correct predictions.
- Off-diagonal values represent misclassifications.

A higher concentration of values along the diagonal indicates better model performance.

Conclusion

The Decision Tree model successfully classified Iris flowers with high accuracy. The confusion matrix demonstrated that most predictions were correct, making the model suitable for this classification task.